Applied Regression Interpretation

Gov 51: Data Analysis and Politics

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The Big Picture

Women earn 84 cents for every dollar men earn

Where does that number come from?

And what does it actually mean?

Regression turns data into claims you can interrogate

- **Today:** One dataset, three models, building complexity
- **Skills you need:**
  - Interpret every coefficient in context
  - Compute $\hat{Y}$ by plug-in
  - Compare models side-by-side
  - Read interactions
- **Dataset:** 526 workers, 1976 Current Population Survey

The Wooldridge wage dataset has six variables we need

Variable	Type	Description
`wage`	Continuous	Hourly wage ($/hr)
`educ`	Continuous	Years of education
`exper`	Continuous	Years of work experience
`tenure`	Continuous	Years at current job
`female`	Binary (0/1)	= 1 if woman
`married`	Binary (0/1)	= 1 if married

$n = 526$ workers · 274 men, 252 women

Model 1: The Raw Gap

Model 1 asks one question: do men and women earn different wages?

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{female}_i$$

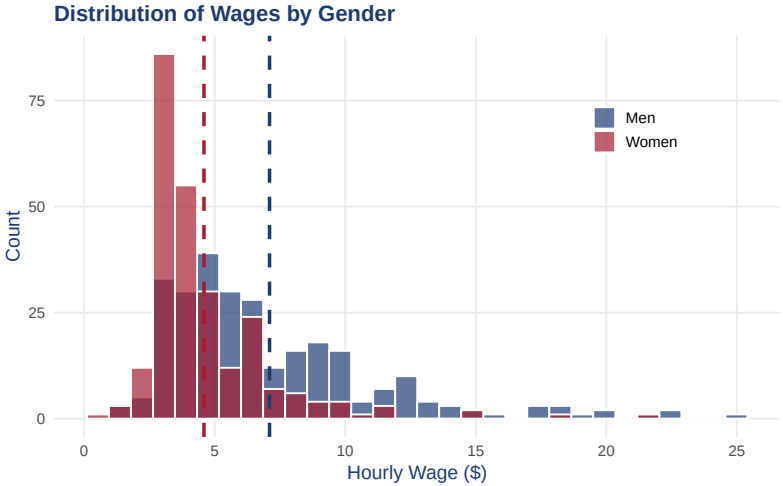
- Binary $X \Rightarrow$ intercept is the mean for the baseline group
- $\hat{\beta}_0 = \overline{\text{wage}}_{\text{men}}$
- $\hat{\beta}_1 = \overline{\text{wage}}_{\text{women}} - \overline{\text{wage}}_{\text{men}}$

Women earn \$2.51 less per hour than men on average

	Model 1
Intercept	7.10
	(0.21)
Female	**−2.51**
	(0.30)
R^2	0.116
n	526

- $\hat{Y}_{\text{men}} = 7.10 \quad \hat{Y}_{\text{women}} = 7.10 - 2.51 = 4.59$
- $t = -8.28 \quad \Rightarrow$ **highly significant**

The wage distribution differs sharply by gender



$R^2 = 0.116$ means gender alone explains 12% of wage variation

- 88% of variation is *within* gender groups
- Some men earn very little; some women earn a lot
- Gender matters, but it is far from the whole story

$$R^2 = 1 - \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\sum \hat{e}_i^2}{\sum (Y_i - \bar{Y})^2} = 0.116$$

Turn to your neighbor

"The wage gap is \$2.51."

What does that number **not** tell you?

Model 2: Adding Covariates

Model 2 adds education, experience, and tenure

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{educ}_i + \hat{\beta}_3 \text{exper}_i + \hat{\beta}_4 \text{tenure}_i$$

- Same outcome, same treatment variable
- Now we **hold constant** education, experience, and tenure
- This asks a *different question* than Model 1

The gender gap shrinks from \$2.51 to \$1.81 after adding controls

	Model 1	Model 2
Intercept	7.10	−1.57
	(0.21)	(0.72)
Female	−2.51	**−1.81**
	(0.30)	(0.26)
Education		0.57
		(0.05)
Experience		0.03
		(0.01)
Tenure		0.14
		(0.02)
R^2	0.116	**0.364**
n	526	526

Every coefficient has a "holding constant" interpretation

- **Female:** $-\$1.81$/hr, holding education, experience, and tenure constant
- **Education:** $+\$0.57$/hr per additional year, holding gender, experience, and tenure constant
- **Experience:** $+\$0.03$/hr per additional year, holding gender, education, and tenure constant
- **Tenure:** $+\$0.14$/hr per additional year, holding gender, education, and experience constant

"Holding constant" changes the question we are asking

Model 1

$\hat{\beta}_{\text{female}} = -2.51$

Gap between the *average* man and the *average* woman

Model 2

$\hat{\beta}_{\text{female}} = -1.81$

Gap between a man and woman with the *same* education, experience, and tenure

Adding covariates changes what $\hat{\beta}_1$ means

R^2 tripled — the model explains 3× more variation

- Model 1: $R^2 = 0.116$
- Model 2: $R^2 = 0.364$
- Education, experience, and tenure explain a lot of wage variation
- The SE on Female *shrank*: $0.30 \rightarrow 0.26$
  - More precise estimate when the model fits better

Turn to your neighbor: plug-in prediction

Using Model 2:

$$\hat{Y} = -1.57 + (-1.81)\text{female} + 0.57 \cdot \text{educ} + 0.03 \cdot \text{exper} + 0.14 \cdot \text{tenure}$$

Predict hourly wage for:

Profile: educ = 16, exper = 10, tenure = 5

1. A woman with this profile: $\hat{Y} = ?$

2. A man with this profile: $\hat{Y} = ?$

3. What is the gap?

Plug-in prediction: the gap is always $\hat{\beta}_{\text{female}}$

- **Man** (female $= 0$):
  $\hat{Y} = -1.57 + 0(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$8.54$
- **Woman** (female $= 1$):
  $\hat{Y} = -1.57 + 1(-1.81) + 0.57(16) + 0.03(10) + 0.14(5) = \$6.72$
- **Gap**: $6.72 - 8.54 = -\$1.81 = \hat{\beta}_{\text{female}}$

With no interaction, the gap is the same for every profile

Model 3: Interactions

Model 3 asks: does marriage affect wages differently for men and women?

$$\widehat{\text{wage}}_i = \hat{\beta}_0 + \hat{\beta}_1 \text{female}_i + \hat{\beta}_2 \text{married}_i + \hat{\beta}_3 (\text{female} \times \text{married})_i + \cdots$$

- $\hat{\beta}_2$ = effect of marriage **for men** (when female = 0)
- $\hat{\beta}_2 + \hat{\beta}_3$ = effect of marriage **for women**
- $\hat{\beta}_3$ = *difference* in marriage effect across genders

Marriage boosts men's wages by \$1.82 but *reduces* women's

	Model 1	Model 2	Model 3
Intercept	7.10	−1.57	−2.39
	(0.21)	(0.72)	(0.73)
Female	−2.51	−1.81	−0.31
	(0.30)	(0.26)	(0.41)
Married			1.82
			(0.40)
Female × Married			**−2.40**
			(0.53)
Education		0.57	0.55
		(0.05)	(0.05)
Experience		0.03	0.02
		(0.01)	(0.01)
Tenure		0.14	0.13
		(0.02)	(0.02)
R^2	0.116	0.364	**0.392**
n	526	526	526

The marriage premium: \$1.82 for men, −\$0.57 for women

Men (female = 0)

Marriage effect = $\hat{\beta}_2$
= +\$1.82/hr

Married men earn significantly more

Women (female = 1)

Marriage effect = $\hat{\beta}_2 + \hat{\beta}_3$
= $1.82 + (-2.40) = -\$0.57$/hr

Married women earn slightly less

$\hat{\beta}_3 = -2.40$ captures the *difference* in marriage premiums

Turn to your neighbor: 2×2 predictions

Using Model 3 with educ = 12, exper = 10, tenure = 5:

$$\hat{Y} = -2.39 + (-0.31)\text{female} + 1.82 \cdot \text{married} + (-2.40)\text{female} \times \text{married}$$
$$+0.55 \cdot \text{educ} + 0.02 \cdot \text{exper} + 0.13 \cdot \text{tenure}$$

Fill in the table:

	Single	**Married**
Men	?	?
Women	?	?

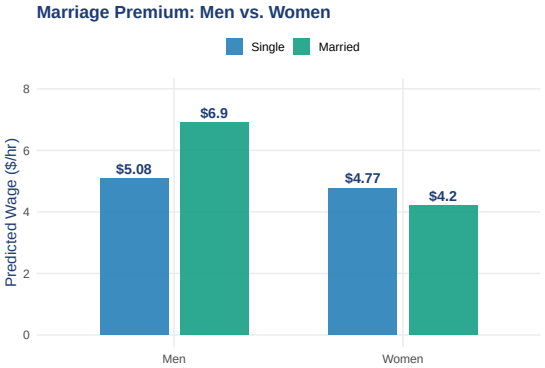
Which group earns the most? The least?

Married men earn the most; married women earn the least

	Single	Married
Men	$5.08	$6.90
Women	$4.77	$4.20

Marriage premium:

Men: +$1.82 Women: −$0.57



Putting It All Together

Track $\hat{\beta}_{\text{female}}$ across all three models

	Model 1	Model 2	Model 3
$\hat{\beta}_{\text{female}}$	-2.51	-1.81	-0.31
	(0.30)	(0.26)	(0.41)
Interpretation	Raw gap	Adjusted gap (same human capital)	Gap for single workers (same HC)
R^2	0.116	0.364	0.392

The "wage gap" depends on which question you ask

Four skills to master before the exam

1. **Interpret coefficients in context**
   "One more year of education $\rightarrow$ +\$0.57/hr, holding ... constant"
2. **Compute $\hat{Y}$ by plug-in**
   Substitute values into $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \cdots$
3. **Compare models side-by-side**
   Track how coefficients, R^2, and SEs change
4. **Read interactions**
   $\hat{\beta}_3$ = difference in slopes $\rightarrow$ build the 2×2 table

Exam 1 is Thursday, March 12

- **Review guide:** posted on course website
- **Cheat sheet:** 2 pages, front and back, handwritten or typed
- **Practice:**
  - Plug-in predictions — get fast at arithmetic
  - Interpret every number in a regression table
  - Build 2×2 tables from interaction models

You have all the tools. Practice using them.